

## LHASA, China

WMO: 555910

Lat: 29.6586N

Lon: 91.1353E

Elev: 3650

StdP: 64.50

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |              |            |            | Coldest Month WS/MCDB |            |            |            | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|------------|--------------|------------|------------|-----------------------|------------|------------|------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |            | 99%          |            |            | 0.4%                  |            | 1%         |            |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP           | HR         | MCDB       | WS                    | MCDB       | WS         | MCDB       | MCWS                  | PCWD      |              |
| (a)<br>1      | (b)<br>-8.3 | (c)<br>-7.1 | (d)<br>-31.2                  | (e)<br>0.3 | (f)<br>6.3 | (g)<br>-29.0 | (h)<br>0.4 | (i)<br>5.4 | (j)<br>7.8            | (k)<br>8.8 | (l)<br>6.2 | (m)<br>7.8 | (n)<br>1.5            | (o)<br>90 | (p)<br>0.489 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>6      | (b)<br>10.7            | (c)<br>25.6     | (d)<br>10.6 | (e)<br>24.2 | (f)<br>10.6 | (g)<br>23.1 | (h)<br>10.5 | (i)<br>13.1         | (j)<br>20.7 | (k)<br>12.6 | (l)<br>20.1 | (m)<br>12.2 | (n)<br>19.6 | (o)<br>2.4           | (p)<br>290 |

| Dehumidification DP/MCDB and HR |             |             |             |             |             |            |             |             | Enthalpy/MCDB |             |             |             |             |             | Extreme<br>Max WB |
|---------------------------------|-------------|-------------|-------------|-------------|-------------|------------|-------------|-------------|---------------|-------------|-------------|-------------|-------------|-------------|-------------------|
| 0.4%                            |             |             | 1%          |             |             | 2%         |             |             | 0.4%          |             | 1%          |             | 2%          |             |                   |
| DP                              | HR          | MCDB        | DP          | HR          | MCDB        | DP         | HR          | MCDB        | Enth          | MCDB        | Enth        | MCDB        | Enth        | MCDB        |                   |
| (a)<br>10.7                     | (b)<br>12.7 | (c)<br>15.0 | (d)<br>10.2 | (e)<br>12.3 | (f)<br>14.7 | (g)<br>9.8 | (h)<br>11.9 | (i)<br>14.6 | (j)<br>50.5   | (k)<br>20.9 | (l)<br>48.9 | (m)<br>20.2 | (n)<br>47.5 | (o)<br>19.8 | (p)<br>19.0       |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| 5.7               | 4.8  | 4.2 | -11.2                      | 27.4 | 1.6                | 1.3 | -12.4                                              | 28.3 | -13.3      | 29.1 | -14.2      | 29.9 | -15.4      | 30.8 |  |  |
|                   |      |     | -13.7                      | 14.7 | 0.9                | 1.9 | -14.4                                              | 16.1 | -14.9      | 17.2 | -15.4      | 18.2 | -16.1      | 19.6 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 9.7    | 0.7   | 3.1   | 6.2   | 9.4   | 13.7  | 17.2  | 17.0  | 16.3  | 14.9  | 10.5  | 5.4   | 1.8   |
|                                                                        | DBStd       | 6.41   | 2.76  | 2.86  | 2.69  | 2.66  | 3.11  | 2.68  | 2.13  | 1.79  | 1.94  | 2.81  | 2.18  | 2.58  |
|                                                                        | HDD10.0     | 1071   | 288   | 194   | 123   | 41    | 5     | 0     | 0     | 0     | 0     | 28    | 138   | 253   |
|                                                                        | HDD18.3     | 3178   | 546   | 427   | 377   | 267   | 146   | 52    | 52    | 66    | 104   | 243   | 388   | 512   |
|                                                                        | CDD10.0     | 972    | 0     | 0     | 4     | 24    | 121   | 217   | 218   | 196   | 147   | 44    | 1     | 0     |
|                                                                        | CDD18.3     | 38     | 0     | 0     | 0     | 0     | 3     | 19    | 12    | 3     | 1     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 222    | 0     | 0     | 0     | 1     | 25    | 115   | 64    | 15    | 3     | 0     | 0     | 0     |
|                                                                        | CDH26.7     | 12     | 0     | 0     | 0     | 0     | 1     | 7     | 4     | 0     | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 1.8    | 1.8   | 2.0   | 2.1   | 2.0   | 2.1   | 2.1   | 1.7   | 1.6   | 1.6   | 1.6   | 1.5   | 1.5   |
| (15) Precipitation                                                     | PrecAvg     | 440    | 1     | 1     | 3     | 7     | 27    | 78    | 123   | 130   | 62    | 9     | 1     | 1     |
|                                                                        | PrecMax     | 795    | 5     | 6     | 19    | 27    | 78    | 193   | 209   | 283   | 136   | 37    | 22    | 13    |
|                                                                        | PrecMin     | 229    | 0     | 0     | 0     | 0     | 0     | 6     | 35    | 29    | 15    | 0     | 0     | 0     |
|                                                                        | PrecStd     | 111    | 1     | 1     | 4     | 6     | 21    | 42    | 44    | 51    | 26    | 10    | 3     | 3     |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 15.1  | 15.8  | 19.6  | 22.1  | 25.9  | 27.7  | 27.2  | 24.8  | 23.7  | 21.7  | 17.1  | 14.6  |
|                                                                        |             | MCWB   | 1.1   | 1.6   | 3.6   | 6.1   | 8.9   | 10.6  | 11.2  | 11.9  | 10.6  | 7.6   | 2.8   | 1.2   |
|                                                                        | 2%          | DB     | 11.5  | 13.5  | 16.5  | 19.4  | 23.7  | 26.1  | 25.0  | 23.5  | 22.3  | 20.0  | 15.0  | 12.1  |
|                                                                        |             | MCWB   | -0.3  | 0.5   | 2.4   | 4.7   | 8.2   | 10.4  | 11.5  | 11.5  | 10.2  | 7.1   | 2.1   | 0.0   |
|                                                                        | 5%          | DB     | 9.2   | 11.8  | 14.5  | 17.7  | 21.9  | 24.6  | 23.6  | 22.4  | 21.0  | 18.6  | 13.4  | 10.2  |
|                                                                        |             | MCWB   | -1.5  | -0.3  | 1.5   | 3.9   | 7.5   | 10.4  | 11.6  | 11.4  | 10.1  | 6.5   | 1.4   | -0.8  |
|                                                                        | 10%         | DB     | 7.3   | 10.0  | 12.8  | 16.1  | 20.2  | 23.2  | 22.2  | 21.2  | 19.8  | 17.0  | 11.8  | 8.3   |
|                                                                        |             | MCWB   | -2.4  | -1.0  | 0.8   | 3.5   | 7.0   | 10.3  | 11.5  | 11.3  | 9.8   | 5.7   | 0.7   | -1.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 1.7   | 2.1   | 4.2   | 7.0   | 11.4  | 13.3  | 14.0  | 13.7  | 12.5  | 9.6   | 4.3   | 1.5   |
|                                                                        |             | MCDB   | 14.3  | 14.4  | 17.3  | 17.8  | 21.6  | 21.8  | 21.5  | 21.1  | 20.0  | 17.9  | 13.8  | 13.6  |
|                                                                        | 2%          | WB     | -0.1  | 0.8   | 2.9   | 5.9   | 9.7   | 12.5  | 13.2  | 13.0  | 11.7  | 8.6   | 2.9   | 0.1   |
|                                                                        |             | MCDB   | 10.8  | 12.4  | 14.4  | 16.5  | 19.9  | 21.1  | 20.7  | 20.4  | 18.9  | 17.0  | 12.8  | 11.3  |
|                                                                        | 5%          | WB     | -1.3  | 0.0   | 2.0   | 5.1   | 8.8   | 11.9  | 12.7  | 12.5  | 11.2  | 7.9   | 1.9   | -0.7  |
|                                                                        |             | MCDB   | 8.7   | 11.0  | 12.7  | 15.1  | 18.7  | 20.5  | 20.1  | 19.6  | 18.1  | 15.5  | 11.9  | 9.7   |
|                                                                        | 10%         | WB     | -2.3  | -0.8  | 1.3   | 4.3   | 8.1   | 11.4  | 12.3  | 12.1  | 10.7  | 7.1   | 1.1   | -1.6  |
|                                                                        |             | MCDB   | 6.8   | 9.4   | 11.4  | 13.5  | 17.6  | 19.7  | 19.6  | 18.9  | 17.4  | 14.5  | 10.8  | 8.0   |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 13.6  | 12.8  | 12.2  | 11.8  | 11.3  | 10.7  | 9.9   | 9.7   | 10.3  | 12.8  | 14.4  | 14.5  |
|                                                                        | 5% DB       | MCDBR  | 16.8  | 15.6  | 15.0  | 14.2  | 12.9  | 11.9  | 11.4  | 11.1  | 11.7  | 14.2  | 16.3  | 16.7  |
|                                                                        |             | MCWBR  | 8.9   | 7.9   | 6.5   | 5.0   | 3.7   | 3.1   | 2.9   | 2.8   | 3.1   | 4.6   | 7.2   | 8.5   |
|                                                                        | 5% WB       | MCDBR  | 16.3  | 15.0  | 13.8  | 12.6  | 11.8  | 10.5  | 9.9   | 9.7   | 9.9   | 11.8  | 15.5  | 16.3  |
|                                                                        |             | MCWBR  | 8.8   | 7.7   | 6.0   | 4.5   | 3.7   | 3.3   | 3.2   | 3.1   | 3.2   | 3.9   | 6.9   | 8.4   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.193 | 0.215 | 0.263 | 0.315 | 0.314 | 0.305 | 0.305 | 0.302 | 0.283 | 0.243 | 0.202 | 0.184 |
|                                                                        | taud        |        | 2.679 | 2.608 | 2.469 | 2.337 | 2.424 | 2.535 | 2.577 | 2.578 | 2.597 | 2.676 | 2.732 | 2.769 |
|                                                                        | Ebn at Noon |        | 1089  | 1078  | 1030  | 977   | 972   | 974   | 973   | 978   | 995   | 1029  | 1069  | 1092  |
|                                                                        | Edh at Noon |        | 73    | 86    | 107   | 127   | 117   | 105   | 100   | 99    | 93    | 80    | 68    | 63    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 4.32  | 5.08  | 5.79  | 6.16  | 6.43  | 6.53  | 5.78  | 5.54  | 5.37  | 5.25  | 4.70  | 4.23  |
|                                                                        | RadStd      |        | 0.29  | 0.32  | 0.38  | 0.37  | 0.32  | 0.35  | 0.36  | 0.33  | 0.30  | 0.31  | 0.26  | 0.17  |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                       |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only          | +0.56 | +0.56   | -3.43  | N/A     | N/A   | -0.37 | -132        | -254    | +150    | +18     |
| Regional (1 neighbor) | +0.35 | N/A     | -3.42  | N/A     | N/A   | N/A   | -132        | -254    | +150    | +18     |

Nomenclature: See separate page